

IT-324 System Integration and Architecture

Final Project Specification.

To effectively accomplish our project, we have identified seven subsystems and assigned each group to an office. A group is also required to identify another office where data will be shared with.

To ensure a smooth and organized workflow, we will follow the project guide outlined below:



Scope definition

1. Identify Project Objective

- What problem are you trying to solve for the chosen office?
- Identify proof to demonstrate

Examples:

- To enhance the University Library's resources and services to better support research and education.
- To Streamline the library's book borrowing process.

2. Gather Stakeholder Input

- What are the students' needs and pain points related to your objective? Shortlist and identify the most important need and a corresponding pain point.

3. Deliverables

- What tangible outcomes will address the objective?

Examples:

- Implement a new online catalog system with advanced search functionality.
- Develop a mobile app for library book borrowing and return.

4. Establish Boundaries

What functionalities or aspects will NOT be part of the project?

Example:

The mobile app does NOT include registration process.

5. Document Scope Statement

- Write a concise scope statement that summarizes what the project will accomplish, including its objectives, deliverables, boundaries, and constraints. This statement serves as a reference point throughout the project to ensure everyone stays aligned.

Dependency mapping

1. Identify dependencies between systems and tasks.
 - Identify which tasks must be completed before others can begin, ensuring a logical flow and preventing roadblocks.
2. Understand Data Flow
 - Determine how data flows between the chosen offices. This involves understanding what data is exchanged, how often it is exchanged, and in what format.
3. List dependencies

- Identify upstream dependencies (data flowing OUT from the office)
- Identify downstream dependencies (data flowing INTO an office)

Example:

Upstream dependency: The library catalog system depends on the student database in the registrar's office to update borrower information.

Downstream Dependency: The student database in the registrar's office depends on the library catalog system to update student records with information about borrowed books.

4. Identify critical dependencies

- Determine which dependencies are critical to the operation of both office

5. Document dependency mapping

- Write a summary document that outlines the dependencies mapped between the two offices. Include descriptions of each dependency, its importance, and any potential risks associated with it. This document serves as a reference for understanding the data integration between the offices and can help in identifying areas for improvement or mitigation strategies.

Interface Management

1. Identify exchange needs

- Database entities and fields from each side

2. Define data formats

- Decide on the format in which data will be exchanged between the two offices. Common data formats include CSV, XML, JSON. Justify the choice of data format through a case situation between your chosen office to the identified office (of choice).

3. Select communication protocols

- Choose the communication protocols that will be used for transmitting data between the offices, protocols such as HTTP, FTP, SOAP, REST, or even direct database connections. Justify the

choice of protocol through a case situation between your chosen office to the identified office (of choice). Include security measures.

4. Document Interface Specifications:

- Create a detailed interface specification document that outlines all aspects of the data exchange process including Data formats and Communication protocols.

Report Formatting Instructions

1. Paper Size: 8.5 inches x 11 inches (standard letter size)
2. Margins: Use 1-inch margins on all sides of the page (top, bottom, left, and right).
3. Font and line spacing:
 - a. Use a legible font such as 11-point Calibri, 11-point Arial, 12-point Times New Roman, or 11-point Georgia. The default font of your word-processing program is acceptable.
 - b. Line Spacing: Double-spaced throughout the entire report, including in-text citations and the reference list.
 - c. Do not add extra spacing between paragraphs.
4. Page Numbering:
 - a. Start page numbering on the first page of the report body (page 2 after the cover page).
 - b. Page numbers should be placed in the top right corner of each page.
5. Exclusions:

- a. The cover page and reference list are not included in the 5-page requirement.
- b. Referencing (APA Format)
- c. Use the American Psychological Association (APA) style, 7th edition, for in-text citations and the reference list.
- d. You are required to have a minimum of 15 and a maximum of 20 references.
- e. Ensure your references are credible and relevant to your report topic

6. Submission

- Online: PDF, Google link to be submitted in My Virtual
- In-class: Hardcopy (No stapler, no folder, Use double clip)